Chapter 19, Problem 18PP

Problem

Realize the following transfer function using an *LC* ladder network terminated in a **1-Ω** resistor:

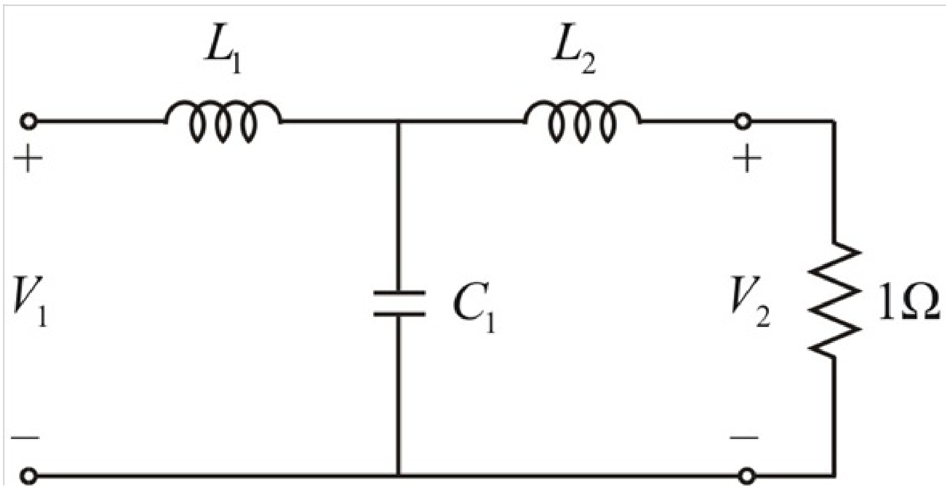
$$H(s) = \frac{2}{s^3 + s^2 + 4s + 2}$$

Step-by-step solution

Step 1 of 8

The denominator of the given transfer function is a third – order network, so that the *LC* ladder network with two inductors and one capacitor.

Step 2 of 8



Step 3 of 8

The terms in the denominator into odd or even groups, then

$$D(s)=(s^3+4s)+(s^2+2)$$

Hence

$$H(s)=\frac{2}{(s^3+4s)+(s^2+2)}$$

Divide the numerator and denominator by the odd parts of the denominator, we get

$$H(s)=\frac{\frac{2}{s^3+4s}}{1+\frac{s^2+2}{s^3+4s}}$$ ----- (1)

By the admittance parameters

$$H(s)=\frac{-y_{21}}{y_L+y_{22}}$$ ----- (2)

Where $y_L=1$ by comparing equations (1), (2), we get

$$y_{21}=\frac{-2}{s^3+4s}$$

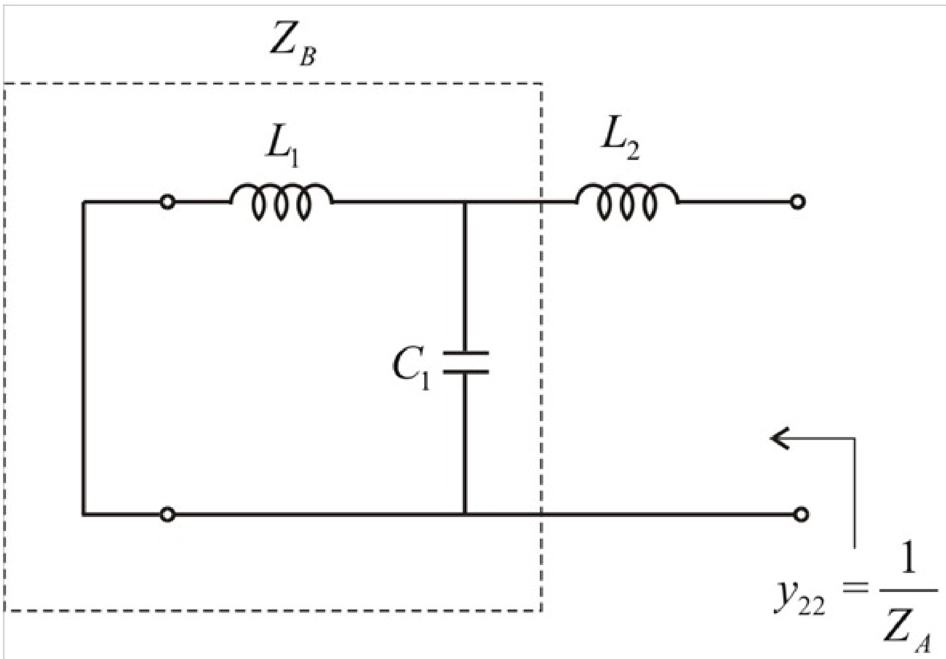
$$y_{22}=\frac{s^2+2}{s^3+4s}$$

Step 4 of 8

y_{22} will automatically realize y_{21}, y_{22} is the output driving – point admittance, the admittance of the network with the input port short – circuited. Thus, y_{22} is the short – circuit output admittance

We short – circuit the input port then circuit is

Step 5 of 8



Step 6 of 8

To get L_3, let

$$Z_A = \frac{1}{y_{22}}$$

$$Z_A = \frac{s^3+4s}{s^2+2}$$

$$Z_A = \frac{s\left(s^2+4\right)}{s^2+2}$$

We get $Z_A = sL_2 + Z_B$ ----- (3)

Thus, by Long division method,

$$\begin{array}{r l}
s^2+2)\ s^3+4s & (s \\
s^3+2s & \\
\hline
2s &
\end{array}$$

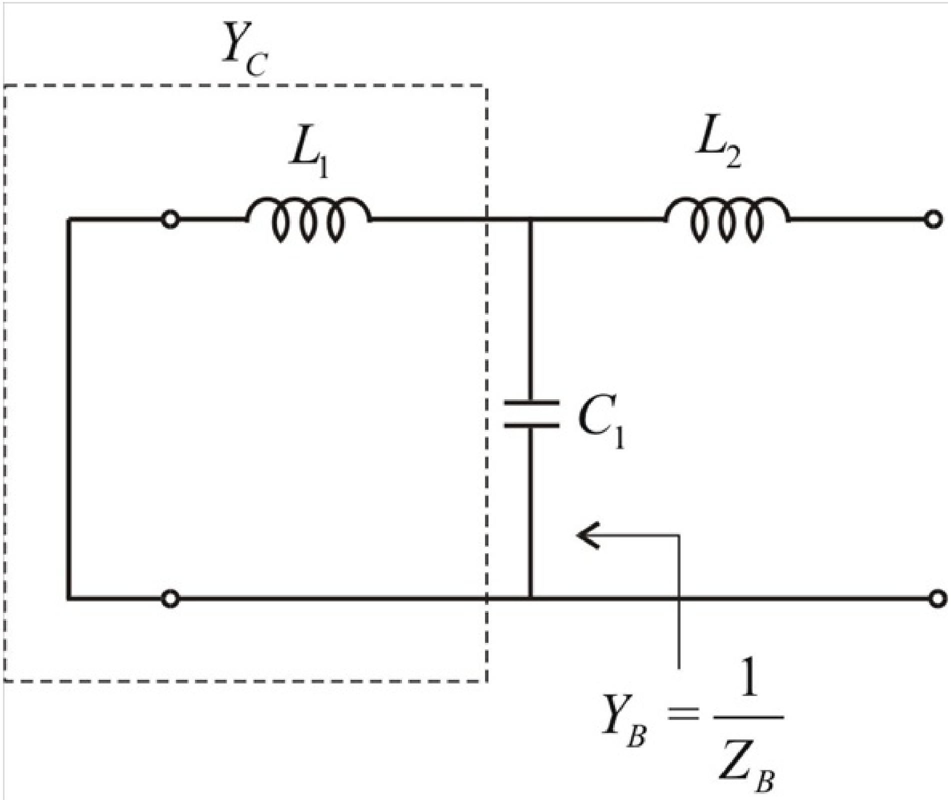
Hence $Z_A = s + \dfrac{2s}{s^2+2}$ ----- (4)

Comparing equation (3) and (4), we get

$\therefore$ $\boxed{L_2 = 1\text{H}}$

$$Z_B = \frac{2s}{s^2+2}$$

Step 7 of 8



Step 8 of 8

Let $Y_B = \frac{1}{Z_B}$

$$Y_B = \frac{s^2+2}{2s}$$

$Y_B = \frac{s}{2} + \frac{1}{s}$ ----- (5)

$Y_B = sC_1 + Y_C$ ----- (6)

By comparing (5) and (6), we get

$$\boxed{\therefore\ C_1 = 0.5\ \text{F}}$$

$$Y_C = \frac{1}{s}$$

$$Y_C = \frac{1}{sL_1}$$

Then $\boxed{L_1 = 1\text{H}}$

Thus, the *LC* ladder network with $L_1 = L_2 = 1\text{H}$, $C_1 = 0.5\ \text{F}$